



COLLEGE OF ENGINEERING & TECHNOLOGY

Department : Computer Engineering

Lecturer : Prof. Gouda Ismail

Course Name : Computer Graphics

Course Code : CC416

Date : 10/6/2024

Time allowed : Two Hours

Start time : 12:00

Final Examination paper

Answer the following questions:

Question 1: (10 points)

- a) Compare between the two basic types of graphics display: Raster scan and Random Scan.
- b) If we have two endpoints (1, 4) and (6, 8), using Bresenham's raster line drawing algorithm, show the set of points we can calculate between the two endpoints to draw the line, and then draw the diagram with the line.
- c) Find the coordinates of point R, if

The transformation matrix $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ maps point P to point Q.

The transformation matrix $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ maps point Q to point R.

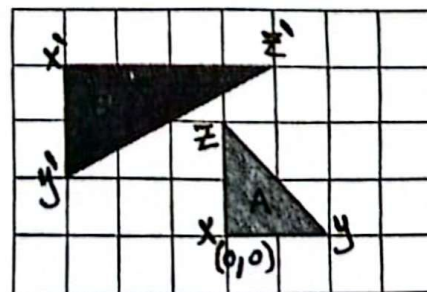
Point P is (-3, 2).

- d) Find the values of x and y, if

The matrix $\begin{pmatrix} 3 & x \\ 6 & -3 \end{pmatrix}$ maps the point (y, 4) onto the point (-14, 24)

Question 2: (8 points)

- a) Assume the origin is placed in the lower left corner of triangle A. Find the transformation matrix that transform the triangle A to triangle B





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- b) Similarly to the case with lines, there is an incremental algorithm for drawing circles – the *mid-point circle algorithm*

- 1) Prove the initial decision parameter of the circle is $p_0 = \frac{5}{4} - r$
- 2) Prove the decision parameter at each point is :

If $p_k < 0$:

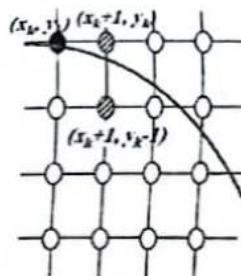
Plot $(x_k + 1, y_k)$

$$p_{k+1} = p_k + 2(x_k + 1) + 1$$

else

Plot $(x_k + 1, y_k - 1)$

$$p_{k+1} = p_k + 2(x_k + 1) + 1 - 2(y_k - 1)$$



Question 3: (12 points)

- a) In 3D, Prove that the *reflection relative to a given axis is equivalent to 180 Degree rotation*?
- b) Consider a triangle whose vertices are (1, 2), (3, 2) and (3, 3). Shear this triangle with $shx=2$ and $shy=1.5$
 - 1) About the origin.
 - 2) About the reference point (3, 5)
- c) A 3-D point A = (3, 4, 5) is reflected relative to XZ-plane, translated in the X-direction by 5, finally scaled along the Y-axis by 2.
 - 1) Construct the *composite transformation matrix* for the above sequence 3-D transformations if all transformations are with respect to the reference frame.
 - 2) Calculate the new position of the point A, if it is transformed by the above sequence 3-D transformations.

Question 4: (10 points)

a) For a rectangular clipping window whose lower left corner is (6,5) and upper right corner is (11,8), Given the following lines AB, and CD Line AB: $A(x_0,y_0) = (6,12)$ and $B(x_1,y_1) = (24,18)$ Line CD: $C(x_0,y_0) = (15,23)$ and $D(x_1,y_1) = (25,22)$	1) <u>Find</u> the region codes for the endpoints of each line 2) <u>Apply</u> Cohen Sutherland algorithm <u>step by step</u> to find the <i>category of each line</i> against the clipping window with showing <i>the new end points of the clipped line</i>												
b) Plot the Bezier curve along with the given <u>three control points</u> .	<table><tr><td></td><td>x</td><td>y</td></tr><tr><td>P₀</td><td>1</td><td>4</td></tr><tr><td>P₁</td><td>2</td><td>7</td></tr><tr><td>P₂</td><td>5</td><td>2</td></tr></table>		x	y	P ₀	1	4	P ₁	2	7	P ₂	5	2
	x	y											
P ₀	1	4											
P ₁	2	7											
P ₂	5	2											

Good Luck